First Progress Review

Chapter 3: Research Methodology

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

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3. Research Methodology

This chapter sets out how the research questions will be answered. It describes the research philosophy and design (Sections 3.1 and 3.2), the framework under study and the escalation rule that is its central mechanism (Sections 3.3 and 3.4), a step-by-step account of how each objective will be achieved (Section 3.5), the populations, samples and data (Sections 3.6 and 3.7), hypotheses, measures and analysis (Sections 3.8 to 3.10), ethics (Section 3.11) and the technology stack and quality assurance (Section 3.12). The methodology follows the accepted proposal; choices of baselines, metrics and statistical tests have been refined using the methodology extracted from the Research Paper Bank (Chapter 2).

3.1 Research Philosophy and Approach

The research adopts a **pragmatist** philosophy: the value of the knowledge produced is judged by whether it works in its intended clinical setting, and methods are chosen for their fitness to each question rather than from a single paradigm (Creswell & Creswell, 2018; Saunders et al., 2019). The approach is predominantly **deductive**—hypotheses derived from the literature are tested on quantitative data—with an **inductive** strand in which clinicians' accounts of trust and escalation behaviour are analysed for themes. The strategy is **design-science research**, because the principal contribution is a constructed artefact, a

working multi-agent clinical AI framework, evaluated against measurable criteria in its intended environment (Hevner et al., 2004).

3.2 Research Design

3.2.1 Design-science research process

The work follows the six activities of the design-science research methodology (Peffers et al., 2007). Table 3.1 maps each activity to this study. Design science is preferred over a purely experimental design because the central claim—that selective autonomy can be made calibrated, measurable and auditable—cannot be demonstrated by benchmarking a model; it requires building the mechanism into a working system and observing it with clinicians.

Table 3.1: Design-science research activities applied to this study.

Activity	In this study	Output
1. Problem identification	Literature review (Chapter 2); clinical pathway analysis with the clinical supervisor	Six gaps; problem statement
2. Objectives of a solution	Objectives O1-O4; hypotheses H1-H6; success criteria	Requirements and evaluation criteria
3. Design and development	Registry, agents, orchestrator, escalation policy, federated protocol (WP1-WP6)	Versioned artefact
4. Demonstration	Retrospective replay of registry cases through the framework	Audit-logged runs
5. Evaluation	Quantitative experiments (WP2-WP5) and multi-reader clinical evaluation (WP7)	Results against H1-H6
6. Communication	Papers, thesis, open benchmark and protocol	Publications, thesis

3.2.2 Mixed-methods evaluation

The evaluation is mixed-methods (Creswell & Creswell, 2018). The dominant **quantitative** strand comprises retrospective and simulated experiments on registry and public data, measuring segmentation, classification, calibration, escalation and federation performance, followed by a multi-reader, multi-case study. The supporting **qualitative** strand comprises semi-structured interviews and free-text feedback from clinicians on trust, usefulness and the acceptability of escalation behaviour. The strands are combined in an **explanatory sequential** arrangement: quantitative results identify where the framework helps or harms, and interviews explain why. Work proceeds in iterative build-and-evaluate cycles closed by clinician feedback; because the artefact evolves, every result is tied to a recorded model version, escalation-policy version, data version and random seed, and no result is carried forward across a change to either without re-evaluation.

3.3 The Framework Under Study

The framework is a team of specialised software agents coordinated by an orchestrator over a shared registry data layer (Figure 3.1). An **agent** is a self-contained component with one bounded job, a defined input and output, and one obligation that distinguishes it from an ordinary module: it must report a calibrated confidence in what it has produced. Six specialist agents cover the pathway: an **Intake Agent** (registration, de-duplication, de-identification and quality control), an **MRI Perception Agent** (detection, segmentation, volume and suggested type), a **Pathology Agent** (slide-level evidence for type and grade with attention maps), a **Diagnosis and Staging Agent** (fusion of imaging, tissue and available markers into a supported WHO 2021 diagnosis, grade and recommendation, degrading gracefully when a source is missing), a **Surgical Planning and Risk-Path Agent** (patient-specific 3D model, structures at risk and a candidate route with its anatomical uncertainty; proof of concept) and a **Follow-up Agent** (longitudinal comparison and surveillance scheduling).

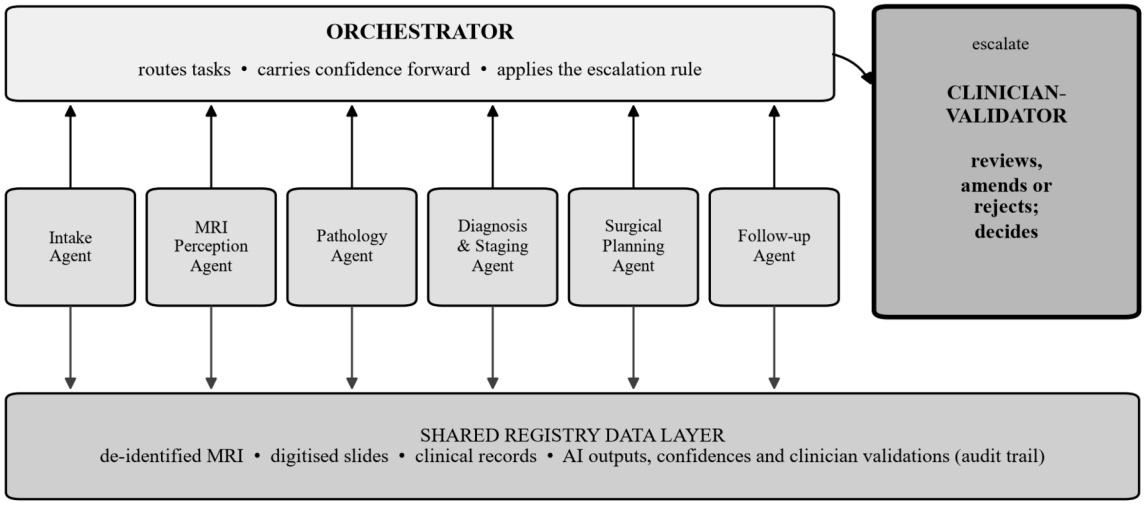


Figure 3.1: System architecture. Specialised agents are coordinated by an orchestrator over a shared registry data layer; every route to a clinical action passes the clinician-validator.

The **orchestrator** is not a diagnostic model. After every agent step it reads the agent's calibrated confidence, combines it with the clinical cost of error for that decision type, applies the escalation rule (Section 3.4) to choose one of three actions—*proceed*, *gather more evidence* or *escalate to a clinician*—and writes the input, output, confidence, action and reason to an append-only audit trail. The **clinician-validator** is a component with a defined role: escalated cases arrive with the agent's output, its evidence and the confidence

that triggered escalation, and the clinician accepts, amends or rejects it. Two invariants hold: there is no route from any agent to a clinical action of record that does not pass a clinician, and every escalation, autonomous step and clinician decision is logged. Figure 3.2 traces one case through the workflow.

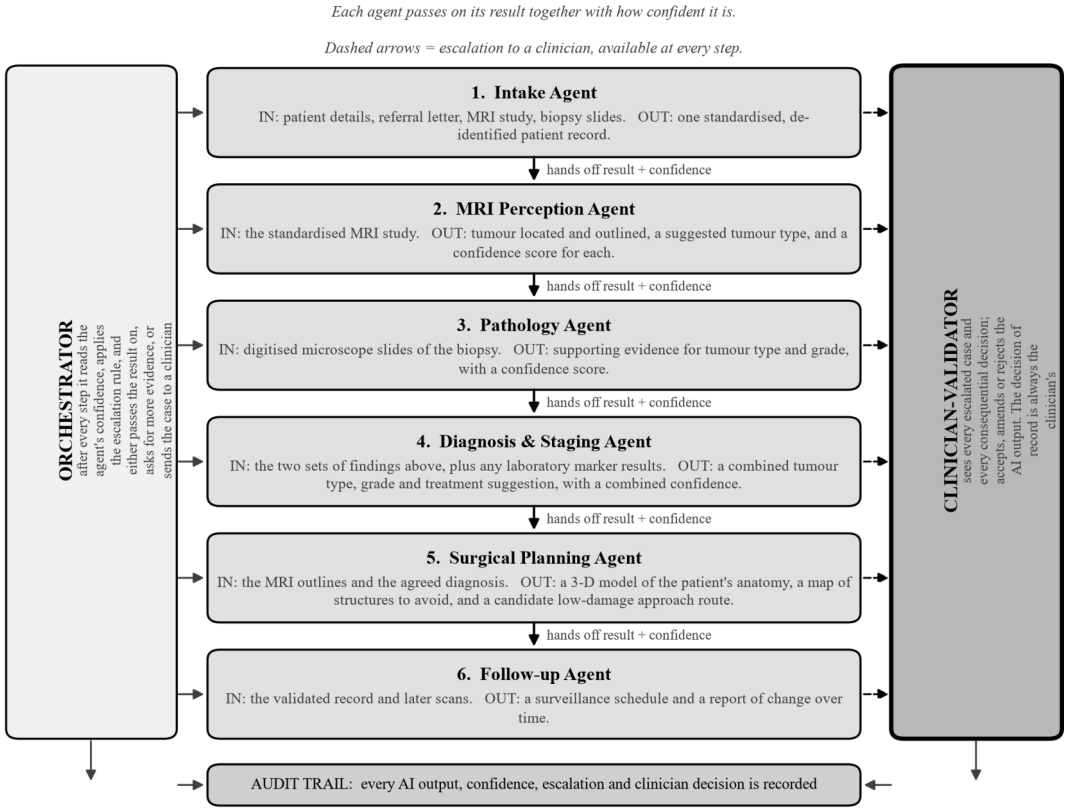


Figure 3.2: The agent workflow from input to final output. The orchestrator applies the escalation rule after every step, and escalation to a clinician is available at every step.

This design responds directly to the literature. Routing by measured, calibrated uncertainty rather than by language-model self-rating addresses the weakness of current medical multi-agent systems (Kim et al., 2024; Lou et al., 2026); isolating tasks in dedicated agents follows evidence that orchestrated designs sustain accuracy under load (Klang et al., 2025); and recording evidence for every claim responds to audits showing that unsupported observations propagate and consensus substitutes for evidence (Zhu et al., 2025). The specialist perception agents are conventional deep-learning models rather than language models, because combining specialist models with a coordinating layer and clinician intervention outperforms either generalist or specialist models alone in modality-specific analysis (Wang et al., 2026).

3.4 The Escalation Rule

The escalation rule is the mechanism under study (Figure 3.3). For agent k and case x, the raw model score is first calibrated so that a reported probability of 0.8 corresponds to being right about 80% of the time; candidate calibrators are temperature scaling (Guo et al., 2017), multi-domain temperature scaling (Yu et al., 2022) and ensemble averaging (Lakshminarayanan et al., 2017). The calibrated probability is converted into an uncertainty $u_k(x)$ (for classification, one minus the maximum calibrated probability or the predictive entropy; for segmentation, an aggregated voxel uncertainty, for which pairwise Dice between ensemble members is a strong baseline (Zenk et al., 2024)).

The threshold is derived from the clinical cost of the two kinds of error rather than chosen by hand. Let c_{FN} be the cost of wrongly reassuring (a missed or under-called finding), c_{FP} the cost of wrongly alarming, and c_R the cost of a clinician review, which reflects scarce specialist time. Acting autonomously on a prediction with calibrated probability p of being correct has expected cost (1 - p) times the cost of the relevant error; escalating costs c_R plus the residual clinician error. The agent therefore proceeds only when

$$(1 - p_k(x)) \cdot c_{err} \leq c_R + \varepsilon_{clin} \cdot c_{err} \quad (3.1)$$

where c_{err} is c_{FN} or c_{FP} according to the error the decision risks and ε_{clin} is the clinician's error rate on that decision. This gives a cost-derived threshold on calibrated confidence for each decision type. A middle band below the threshold triggers *gather more evidence* (for example, waiting for the pathology result before the diagnosis is released), and a list of decisions that are always escalated regardless of confidence (for example, any surgical recommendation, and any conflict between imaging and tissue evidence) is fixed in advance. Out-of-distribution inputs flagged by a separate detector are escalated (Yang et al., 2024). The costs are elicited from the clinical supervisor and participating clinicians through structured scenarios, not assumed, and the policy's sensitivity to them is reported.

Two further mechanisms address the chain of dependent decisions. **Uncertainty propagation:** each agent receives its upstream agents' uncertainty as an input feature and reports a combined uncertainty, so that a doubtful segmentation cannot become a confident grade. **Risk control:** the thresholds are calibrated on held-out data with conformal risk control (Angelopoulos et al., 2024) so that the error rate among autonomously handled cases stays below a target with a stated probability; because such guarantees are marginal and can fail under shift (Mehrtens et al., 2023), they are re-checked at each site and after

every model update. Baselines are a fixed confidence threshold on the uncalibrated softmax (the maximum softmax response, which remains a strong baseline (Bungert et al., 2023)), selective classification with guaranteed risk (Geifman & El-Yaniv, 2017), learning to defer with one-vs-all calibration (Verma & Nalisnick, 2022) and complementarity-driven deferral (Dvijotham et al., 2023).

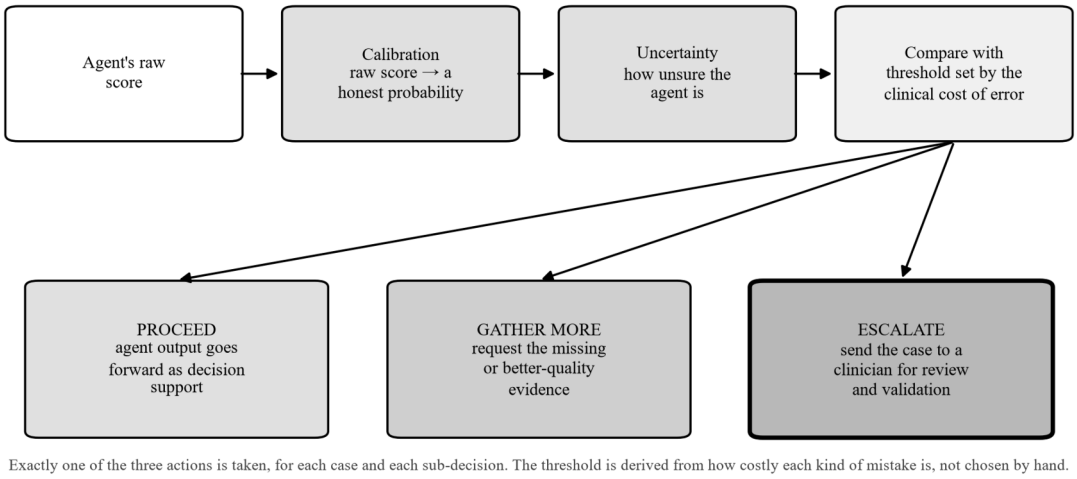


Figure 3.3: How the escalation decision is computed: a raw score is calibrated, converted into an uncertainty, and compared against a threshold derived from the clinical cost of error.

In variable terms, the **independent variables** are the design choices under the researcher's control: uncertainty-estimation method, calibration method, escalation policy and its cost parameters, training regime (local, transfer, self-supervised, federated) and fusion strategy. The **mediating variable** is the calibrated uncertainty attached to each agent output: the thesis claims that the effect of the design choices on outcomes is transmitted through the quality of this estimate rather than through raw accuracy. The **dependent variables** are diagnostic performance, clinician workload, escalation appropriateness, safety and trust. **Moderating variables** recorded for every case are scanner and protocol, image and slide quality, tumour type and grade, completeness of evidence and site.

3.5 How Each Objective Will Be Achieved

This section explains, objective by objective, exactly how the work will be done. Figure 3.4 shows the overall shape of the methodology: four phases over 36 months, each advancing all four objectives, leading to the outputs of the thesis. For every objective the same structure is used: what it must achieve, the research question and hypotheses it tests, its inputs, the step-by-step procedure, the experiments with their baselines and measures,

the outputs, the success criteria and the months in which it is done (from the work plan in Chapter 5). Table 3.2 first summarises the mapping.

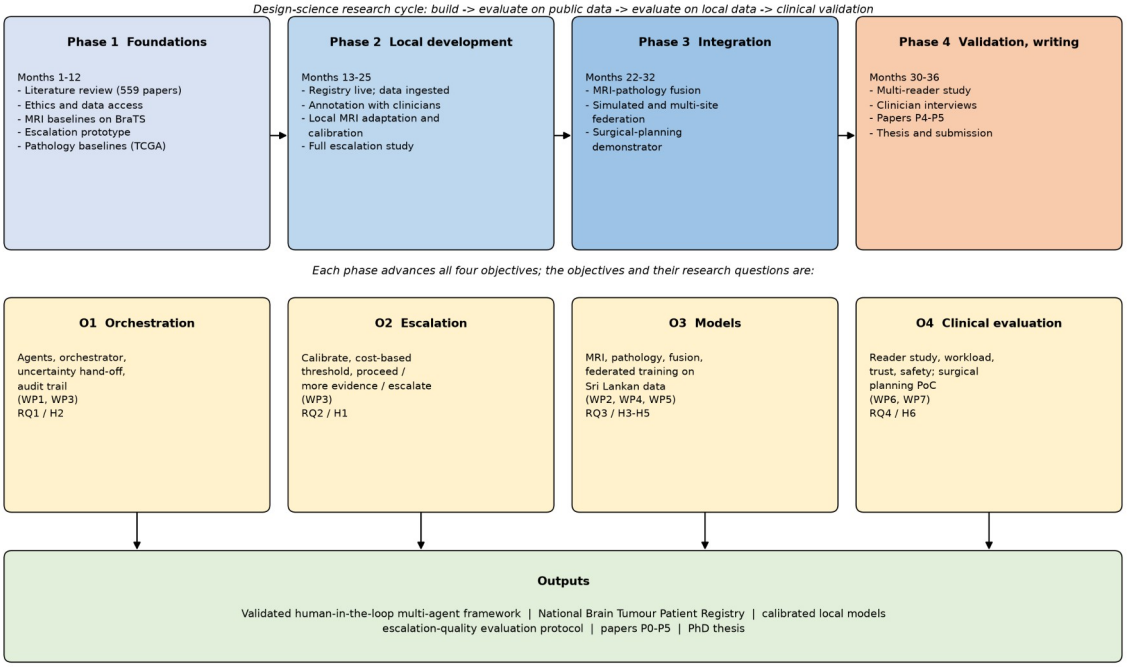


Figure 3.4: The shape of the methodology: four phases of design-science research, the four objectives they advance, and the outputs of the study.

Table 3.2: Objectives, research questions, work packages, data, main methods and outputs.

Objective	RQ / H	Work packages	Data	Main methods	Main output
O1 Orchestration	RQ1 / H2	WP1, WP3	Registry cases (replayed); BraTS while local data accrue	Agent contracts, orchestrator, uncertainty propagation, audit trail	Working multi-agent framework with audit trail
O2 Escalation	RQ2 / H1	WP3	Agent outputs with clinician decisions; elicited costs	Calibration, cost-based thresholds, conformal risk control, learning to defer	Calibrated escalation policy; workload-safety frontier
O3 Models	RQ3 / H3-H5	WP2, WP4, WP5	BraTS, BraTS-Africa, TCGA; Sri Lankan MRI and slides	nnU-Net/Swin UNETR, self-supervised and transfer learning, CLAM, fusion, FedAvg/FedProx/FedBN	Calibrated local models; federated protocol
O4 Clinical evaluation	RQ4 / H6	WP6, WP7	60-100 cases read by 6-8 clinicians; interviews	Multi-reader multi-case study, NASA-TLX, SUS, trust scale, thematic analysis; 3D planning	Clinical evaluation; surgical-planning demonstrator

3.5.1 Objective 1: Multi-agent orchestration with an audit trail

What it must achieve. A working architecture in which six specialist agents are coordinated by an orchestrator, every output carries a calibrated uncertainty across each hand-off, and every AI output and clinician decision is recorded so that any case can be

reconstructed. **Question tested:** RQ1, hypothesis H2 (propagating calibrated uncertainty reduces downstream errors). **Inputs:** the registry (WP1), the agents' models from O3 and the escalation policy from O2. **When:** registry months 5-15; prototype months 7-13 on public data; full study months 19-25 on local data.

Procedure, step by step:

1. **Map the clinical pathway.** With the clinical supervisor, write down every decision point from referral to follow-up (who decides, on what evidence, how long it takes, what an error costs). This becomes the list of agent tasks and decision types.
2. **Write an agent contract for each agent.** For each of the six agents define its input, its output, the uncertainty it must report (a calibrated probability for classes, a voxel-wise uncertainty map plus a case-level score for segmentations) and the conditions under which it was validated (a model-facts statement (Sendak et al., 2020; Mitchell et al., 2019)).
3. **Build the registry and data layer (WP1).** PostgreSQL with an HL7 FHIR resource model (Bender & Sartipi, 2013), an Orthanc DICOM server (Jodogne, 2018), a de-identification and defacing pipeline, automated quality checks and an append-only audit table that stores input, output, uncertainty, action, reason and model/policy version for every step.
4. **Implement the orchestrator.** A rule engine that, after each agent step, reads the uncertainty, applies the escalation rule from O2, chooses proceed, gather more evidence or escalate, routes the case and writes the audit record. Mandatory-review decisions (for example any surgical recommendation, or conflicting imaging and tissue evidence) are coded as fixed rules.
5. **Implement uncertainty propagation.** Each downstream agent receives the upstream agent's output and its uncertainty as inputs; the combined uncertainty is computed (for example by sampling upstream segmentations from the ensemble and passing each through the classifier), so that a doubtful segmentation produces a correspondingly doubtful grade; how uncertainty is communicated to the next agent matters (Hauke et al., 2026).
6. **Build the clinician interface.** A screen showing the case, the agent's output, its evidence (heatmaps, measurements) and the confidence that triggered escalation, with accept, amend and reject buttons that write to the audit trail.

7. **Prototype on public data (months 7-13).** Run BraTS cases through the MRI and classification agents with the orchestrator, using the BraTS labels in place of clinician decisions, to test the plumbing and the propagation method.
8. **Replay registry cases (months 19-25).** Run retrospective local cases through the full chain; the recorded clinician diagnosis acts as the reference decision.
9. **Test H2.** Compare two versions of the chain on the same cases: (a) hand-offs that pass only the output, and (b) hand-offs that pass output plus calibrated uncertainty. Count downstream errors (wrong grade, wrong recommendation) and trace each error back to its first wrong step, following the process-level audit approach of Zhu et al. (2025).
10. **Check auditability.** For a random sample of cases, reconstruct every decision from the audit trail alone and confirm that it matches what happened.

Table 3.3: Experiments for Objective 1.

Experiment	Compared	Data	Measures
E1.1 Error propagation (H2)	Hand-off with vs without uncertainty	Registry replay; BraTS	Downstream error rate, errors traced to origin, McNemar's test
E1.2 Orchestrated vs single model	Six agents vs one end-to-end model	Registry replay	Accuracy, calibration, failure categories
E1.3 Audit completeness	-	Random 10% of cases	Share of decisions fully reconstructable (target 100%)
E1.4 Robustness	Missing or poor input at each step	Simulated corruption	Escalation triggered, errors avoided

Outputs: the framework code, the agent contracts, the registry, and the error-propagation analysis (part of paper P2). **Success criteria:** every route to a clinical action passes a clinician; 100% of sampled decisions reconstructable from the audit trail; a statistically significant reduction in downstream errors with uncertainty propagation (H2).

3.5.2 Objective 2: The calibrated, cost-based escalation mechanism

What it must achieve. A rule that decides, for each agent output, whether the system may proceed, must gather more evidence, or must escalate to a clinician, so that clinician workload falls while errors among autonomously handled cases stay below an agreed level. **Question tested:** RQ2, hypothesis H1. **Inputs:** agent outputs with uncertainties (O1, O3), recorded clinician decisions, and error costs elicited from clinicians. **When:** prototype months 7-13; full study months 19-25; cost elicitation also in months 24-27 for the clinical study.

Procedure, step by step:

11. **Split the data.** For each agent create three patient-level sets: training, calibration (held out, about 20%) and test. The calibration set is never used to train a model.
12. **Estimate uncertainty.** For each model compute three estimators: maximum softmax probability (baseline), Monte Carlo dropout (20 passes) and a 5-member deep ensemble (Gal & Ghahramani, 2016; Lakshminarayanan et al., 2017); for segmentation also test-time augmentation and pairwise Dice between ensemble members (Zenk et al., 2024).
13. **Calibrate.** Fit temperature scaling on the calibration set (Guo et al., 2017), and multi-domain temperature scaling when several sites are present (Yu et al., 2022). Measure expected calibration error, Brier score and reliability diagrams before and after.
14. **Elicit error costs.** With the clinical supervisor and participating clinicians, use structured scenarios for each decision type (for example 'a high-grade glioma reported as low-grade') to rate the harm of a false reassurance, a false alarm and the cost of a specialist review. Record the median and range; the policy is later tested across this range.
15. **Derive the thresholds.** Apply the decision rule (Equation 3.1) to turn the costs into a confidence threshold for each decision type, with a middle 'gather more evidence' band and the fixed list of mandatory escalations.
16. **Add a statistical guarantee.** Use conformal risk control (Angelopoulos et al., 2024) on the calibration set to adjust each threshold so that the error among accepted cases is at most the target (for example 5%) with stated confidence.
17. **Add a novelty check.** Flag out-of-distribution inputs (new scanner, unusual tumour) with a feature-distance detector (Yang et al., 2024) and escalate them regardless of confidence.
18. **Run the baselines.** Fixed threshold on uncalibrated softmax; selective classification with guaranteed risk (Geifman & El-Yaniv, 2017); learning to defer with one-vs-all calibration (Verma & Nalisnick, 2022); complementarity-driven deferral (Dvijotham et al., 2023).
19. **Draw risk-coverage curves.** For every method, vary the threshold and plot selective risk against coverage (share of cases handled without a clinician); compute the area under the curve and the workload-safety frontier, with paired bootstrap intervals (Efron & Tibshirani, 1993).

20. **Test H1 and sensitivity.** Compare methods at matched coverage and at matched risk; repeat after changing the costs within the elicited range and after moving to a new site, to show how stable the policy is.

Table 3.4: Experiments for Objective 2.

Experiment	Compared	Data	Measures
E2.1 Calibration	Raw, temperature, multi-domain temperature, ensemble	Calibration and test sets	ECE, MCE, Brier score, reliability diagrams
E2.2 Escalation policy (H1)	Cost-based + conformal vs four baselines	Local test set; BraTS	Selective risk, coverage, AURC, missed-escalation rate
E2.3 Cost sensitivity	Costs across elicited range	Local test set	Change in coverage and risk
E2.4 Shift	Same policy at a new site / after model update	Leave-one-site-out; BraTS-Africa	Coverage of the guarantee, calibration drift
E2.5 Escalation triggers	Low confidence, novelty, conflict, mandatory	Registry replay	Share and appropriateness of each trigger

Outputs: the escalation policy, the cost-elicitation protocol and the workload-safety analysis (flagship paper P2). **Success criteria:** lower selective risk than the fixed-threshold baseline at the same coverage, or higher coverage at the same risk (H1), with the guaranteed error level holding on the test set and its drift under shift reported.

3.5.3 Objective 3: Data-efficient, multi-source and federated models

What it must achieve. The models the agents rely on—MRI segmentation and classification, slide-level pathology analysis, and their fusion—must work and be well calibrated on small, heterogeneous Sri Lankan data, and must be trainable across hospitals without moving patient data. **Questions tested:** RQ3, hypotheses H3 (self-supervised pre-training), H4 (fusion) and H5 (federation). **When:** MRI on public data months 3-10, on local data months 16-21; pathology on public data months 11-16, local slides and fusion months 22-27; federation simulated months 26-27, multi-site months 28-30.

Part A - MRI models (WP2), step by step:

21. **Prepare data.** Convert DICOM to NIfTI; N4 bias-field correction; co-register T1, T1c, T2 and FLAIR; skull-strip with HD-BET (Isensee et al., 2019); resample to 1 mm; record which sequences are missing.
22. **Train public baselines.** nnU-Net (3D full resolution, five-fold cross-validation) (Isensee et al., 2021) and Swin UNETR (Hatamizadeh et al., 2022) on BraTS 2021 (Baid et al., 2021); test externally on BraTS-Africa (Adewole et al., 2026) to measure the drop under shift.
23. **Pre-train on unlabelled local scans (H3).** Self-supervised pre-training of the Swin UNETR encoder on all de-identified local MRI, labelled or not (Tang et al., 2022).

24. **Fine-tune on labelled local cases** in three regimes: public-only, transfer (public then local fine-tuning) and self-supervised plus fine-tuning (Boyd et al., 2024); train with random sequence dropout so that missing sequences are tolerated.
25. **Add uncertainty and calibration** as in O2, and measure Dice, HD95, sensitivity, calibration and the uncertainty-error correlation.
26. **Failure analysis.** Assign each case in the worst 10% to a category (small lesion, post-operative cavity, artefact, missing sequence, unusual tumour) to locate the model's competence boundary.

Part B - Pathology and fusion (WP4), step by step:

27. **Digitise and prepare slides.** Scan H&E slides at 20x (40x where available); remove metadata; detect tissue; tile at 256 x 256 pixels; normalise stain (Macenko method).
28. **Extract features** with a frozen pathology foundation encoder (UNI) (Chen et al., 2024), so that only a small model has to be trained on local labels.
29. **Train slide-level models** with attention-based multiple-instance learning (CLAM) (Lu et al., 2021) for tumour type and grade, first on TCGA glioma slides and then fine-tuned on local slides; keep the attention maps for the clinician view.
30. **Fuse sources (H4).** Combine the MRI representation, the slide representation and available markers (IDH, 1p/19q, MGMT) with late fusion and a mixture-of-experts model (Saueressig et al., 2026), trained with random source dropout so that it still works when a source is missing.
31. **Compare** imaging alone, tissue alone and their combination against the WHO 2021 reference diagnosis, including simulated missing sources.

Part C - Federated training (WP5), step by step:

32. **Simulate first.** Split the local and public data by site and train with NVIDIA FLARE (Roth et al., 2022) on one machine: FedAvg, FedProx and FedBN (McMahan et al., 2017; Li et al., 2020; Li et al., 2021) against single-site and pooled (centralised) training.
33. **Measure calibration after every round,** and re-check the escalation policy's coverage and risk on each round's model, because a model can gain accuracy while losing calibration.
34. **Add privacy protection** where required (differential privacy with a recorded privacy budget (Abadi et al., 2016)) and measure its cost in accuracy.

35. **Deploy to real sites** (at least two hospitals, if agreements allow): a FLARE client inside each hospital network, only parameters exchanged, with an audit showing no patient data left the site.
36. **Compare (H5):** federated versus single-site (generalisation to the other site) and versus centralised (difference within a pre-defined margin), including calibration.

Table 3.5: Experiments for Objective 3.

Experiment	Compared	Data	Measures
E3.1 MRI regimes (H3)	Public-only vs transfer vs self-supervised + fine-tuning	BraTS, BraTS-Africa, local MRI	Dice, HD95, ECE, Wilcoxon signed-rank
E3.2 Missing sequences	Full vs dropout-trained vs missing-modality model	Local MRI with real gaps	Dice by available sequences
E3.3 Pathology	CLAM with UNI vs ImageNet features	TCGA, local slides	AUROC, balanced accuracy, calibration
E3.4 Fusion (H4)	MRI vs slide vs fusion; missing-source ablation	Local linked cases	Concordance with WHO 2021 diagnosis, DeLong, McNemar
E3.5 Federation (H5)	FedAvg, FedProx, FedBN vs single-site vs pooled	Site-partitioned and real sites	Dice/AUROC difference, ECE change, communication cost

Outputs: calibrated MRI, pathology and fusion models for the agents, the federated protocol, and papers P1, P3 and P4. **Success criteria:** local models clinically useful and calibrated on the held-out local test set (for example whole-tumour Dice within the inter-rater range and ECE below 0.05); fusion better than the best single source (H4); federated models within the pre-defined margin of centralised training with calibration preserved (H5).

3.5.4 Objective 4: Clinical evaluation of the human-in-the-loop framework

What it must achieve. Evidence of whether the framework helps clinicians in practice: diagnostic performance, the appropriateness of its escalations, workload, usability, trust and safety; and a proof of concept showing whether MRI-based surgical-planning support is feasible. **Question tested:** RQ4, hypothesis H6. **When:** protocol and ethics amendment months 24-27; reader study months 30-33; analysis months 33-34; surgical-planning demonstrator months 31-32.

Part A - Multi-reader, multi-case study (WP7), step by step:

37. **Write the protocol** following DECIDE-AI (Vasey et al., 2022); pre-register endpoints (primary: review time and missed-escalation rate; secondary: accuracy, usability, workload, trust) and submit an ethics amendment.

38. **Select cases.** 60-100 retrospective cases not used in development, stratified by tumour type and difficulty, with a confirmed reference diagnosis.
39. **Recruit readers.** 6-8 clinicians (neurosurgeons, radiologists, histopathologists, senior trainees) by purposive sampling; obtain consent.
40. **Train readers** on the interface with a short session and five practice cases not in the study set.
41. **Read in two conditions.** Each reader reads every case unassisted and AI-assisted, in counterbalanced order with a washout of at least four weeks; the interface logs time and every decision.
42. **Blinded safety review.** An independent expert reviews all cases the system handled without escalation and marks any that should have been escalated (the missed-escalation rate).
43. **Questionnaires** after each condition: NASA Task Load Index (Hart & Staveland, 1988), System Usability Scale (Brooke, 1996) and a trust scale.
44. **Interviews.** 30-45 minute semi-structured interviews on trust, escalation behaviour and fit to the workflow; transcribed and analysed by reflexive thematic analysis (Braun & Clarke, 2006).
45. **Analyse** with mixed-effects models (reader and case as random effects) for time and accuracy, non-inferiority testing for accuracy, and triangulate the interview themes with the logged escalation data; the analysis looks specifically for the shift towards missed diagnoses that selective prediction can cause (Jabbour et al., 2025).

Part B - Surgical-planning proof of concept (WP6), step by step:

46. **Build patient-specific 3D models** in 3D Slicer (Fedorov et al., 2012) from the MRI segmentation, with eloquent cortex, major vessels and tracts from diffusion tractography where available.
47. **Compute a candidate route** with a risk-weighted sampling-based planner (Karaman & Frazzoli, 2011), widening the safety margin where segmentation uncertainty is high.
48. **Ask neurosurgeons to rate** 10-20 plans for anatomical plausibility, usefulness and whether the uncertainty display helps, and record geometric measures (distance to structures at risk).

Table 3.6: Experiments for Objective 4.

Experiment	Compared	Data	Measures
E4.1 Reader study (H6)	Unassisted vs AI-assisted HITL	60-100 cases, 6-8 readers	Review time, accuracy (non-inferiority), mixed-effects models
E4.2 Escalation	Escalated vs autonomous	Study cases	Appropriate-escalation and

Experiment	Compared	Data	Measures
appropriateness	cases		missed-escalation rates
E4.3 Workload, usability, trust	Two conditions	Questionnaires	NASA-TLX, SUS, trust scale
E4.4 Clinician experience	-	Interviews	Themes, triangulated with logs
E4.5 Surgical planning	Plans with vs without uncertainty display	10-20 cases	Surgeon ratings, distance to structures at risk

Outputs: the clinical evaluation and the escalation-quality evaluation protocol (paper P5), and the surgical-planning demonstrator. **Success criteria:** shorter review time with accuracy within the non-inferiority margin (H6), a low missed-escalation rate, usability and trust rated acceptable, and neurosurgeons judging the planning support plausible and useful.

3.6 Population and Sample

Patients. The target population is all patients with primary or secondary brain and nervous-system tumours in Sri Lanka (about 700 new cases a year (International Agency for Research on Cancer, 2024)). The accessible population is patients presenting to the participating tertiary units during the study period, recruited by consecutive sampling. Eligible patients are of any age with a suspected or confirmed brain or nervous-system tumour, at least one usable MRI examination in DICOM form, sufficient clinical information to establish the reference standard, and ethics approval with consent or waiver. Records are excluded where imaging is unusable, the reference standard cannot be established, de-identification cannot be completed, or governance requirements are not met. Patients are not excluded for missing tissue, marker or follow-up data; they contribute according to data availability, which allows evaluation under realistic incompleteness. The expected cohort is several hundred patients over the study period.

Sample-size reasoning. For model development, the required sample is estimated with the Riley criteria for prediction models (Riley et al., 2020); for diagnostic accuracy, from the expected sensitivity and the width of its confidence interval (Obuchowski, 1998). As an illustration, estimating a sensitivity of 0.90 with a 95% confidence interval of ±0.07 requires about 71 positive cases for each class of interest. Because local numbers are bounded by incidence, public data are used for pre-training and baselines, and federated training increases the effective sample without pooling data.

Clinicians. Clinicians are recruited by purposive sampling based on expertise. The multi-reader, multi-case design uses the scarce readers efficiently: with about 6-8 readers and 60-

100 cases, each read under both conditions, the study is powered to detect a clinically meaningful reduction in review time and to test non-inferiority of diagnostic accuracy within a pre-specified margin. **Public data** (BraTS 2021, BraTS-Africa, TCGA glioma slides and other open sets) are used for pre-training, benchmarking and external comparison only, never as the sole basis for a clinical claim.

3.7 Data Sources and Collection

Primary data are diagnostic MRI studies in DICOM, digitised biopsy slides, structured clinical information, operative and pathology reports and follow-up records. Secondary data are public benchmarks and national statistics (International Agency for Research on Cancer, 2024; National Cancer Control Programme, 2024). Reference annotations are produced with 3D Slicer (Fedorov et al., 2012) and QuPath (Bankhead et al., 2017) by trained annotators and reviewed by clinicians; a subset is double-annotated to measure inter-rater agreement. Diagnostic labels come from clinician-confirmed records using WHO 2021 terminology (Louis et al., 2021). Collection starts only after ethics approval; in federated experiments only model parameters leave a hospital (Figure 3.4).

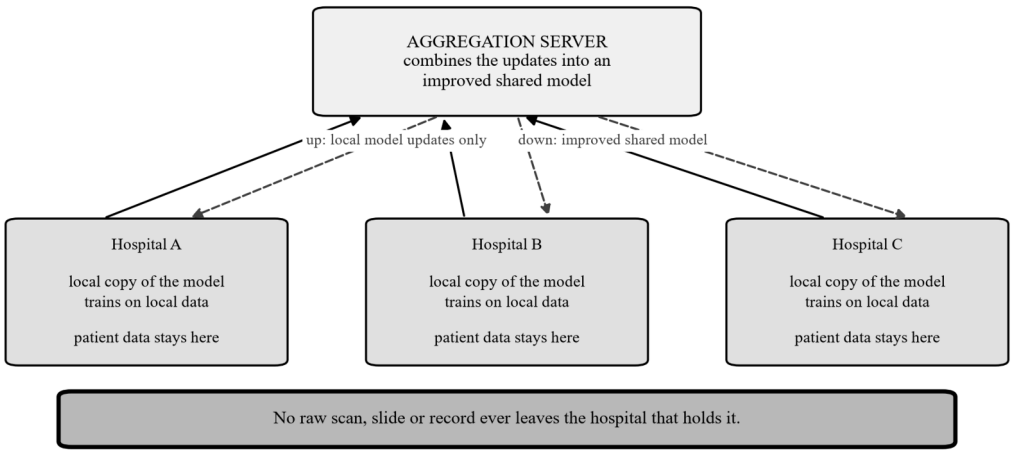


Figure 3.5: Federated training across participating hospitals. Only model parameters are exchanged; raw patient data never leave an institution.

3.8 Hypotheses

The hypotheses are stated with their null form of no difference between compared methods or conditions:

- **H1 (RQ2).** A calibrated, cost-derived escalation policy achieves a better workload-safety trade-off than fixed-confidence thresholding: lower error at the same escalation rate, or lower escalation at the same error rate.
- **H2 (RQ1).** Propagating calibrated uncertainty between agents reduces downstream decision errors compared with hand-offs that ignore upstream reliability.
- **H3 (RQ3).** Self-supervised pre-training on unlabelled local MRI followed by fine-tuning improves performance and calibration compared with transfer from public data alone.
- **H4 (RQ3).** Combining MRI and slide evidence gives higher diagnostic and grading concordance with the reference standard than the best single source.
- **H5 (RQ3).** Federated training improves generalisation over single-site training while staying within a pre-defined margin of centralised training and preserving calibration within a pre-defined tolerance.
- **H6 (RQ4).** The AI-assisted human-in-the-loop workflow reduces clinician review time and improves usability and trust compared with an unassisted workflow, while keeping diagnostic performance within a pre-defined non-inferiority margin.

3.9 Measures

Table 3.7: Operationalisation of the main constructs.

Construct	Measures
Segmentation quality	Dice similarity coefficient, 95th-percentile Hausdorff distance, normalised surface distance, sensitivity and specificity, per case with confidence intervals (Maier-Hein et al., 2024)
Classification and diagnosis	AUROC, balanced accuracy, per-class F1, concordance with the clinician reference standard
Calibration (mediator)	Expected and maximum calibration error (Naeini et al., 2015), Brier score (Brier, 1950), reliability diagrams (Guo et al., 2017), empirical coverage of conformal sets (Angelopoulos & Bates, 2021)
Selective autonomy	Coverage (share handled autonomously), selective risk (error among those), area under the risk-coverage curve (Geifman & El-Yaniv, 2017), appropriate-escalation rate (escalated cases in which the clinician changed the output), missed-escalation rate (autonomous cases judged by blinded experts to need a clinician; primary safety indicator)
Workload	Review minutes per case from the audit log; NASA Task Load Index (Hart & Staveland, 1988)
Usability and trust	System Usability Scale (Brooke, 1996); Likert trust instrument; interview themes
Agreement	Cohen's kappa (Landis & Koch, 1977) and Bland-Altman analysis for continuous measures (Bland & Altman, 1986)
Federation	Federated-minus-centralised performance, calibration change after aggregation, communication cost, privacy budget, audit-verified absence of raw-data transfer

3.10 Data Analysis

Models are developed with patient-level stratified cross-validation and reported once on a held-out local test set untouched during development. Splits are stratified by tumour type and site so that no patient appears in both training and test data; leave-one-site-out evaluation is added where generalisation across sites is the question. For H1 and H2, risk-coverage curves and their areas are compared with paired bootstrap confidence intervals from resampling patients (Efron & Tibshirani, 1993), and the missed-escalation rate at matched coverage is compared with McNemar's test (McNemar, 1947). For H3-H5, paired predictions on the same cases are compared with DeLong's test for AUROC (DeLong et al., 1988), McNemar's test for accuracy and the Wilcoxon signed-rank test for per-case metrics such as Dice. H6 uses mixed-effects models with reader and case as crossed random effects. Family-wise error is controlled within each work package by Holm's procedure (Holm, 1979), with primary endpoints declared in advance. Interviews are analysed by reflexive thematic analysis (Braun & Clarke, 2006) and triangulated with the escalation data. Negative results are reported in the same detail as positive ones, and diagnostic and prediction studies are reported to STARD-AI and TRIPOD+AI (Sounderajah et al., 2025; Collins et al., 2024).

3.11 Ethical, Privacy and Governance Considerations

Ethics approval is obtained before any patient data are accessed. The application to the Research Ethics Committee, Faculty of Medicine, KDU, was finalised with a KDU co-investigator and its processing fee paid on 10 November 2026 for submission; the University of Sri Jayewardenepura application follows KDU approval, and approval from each participating hospital is obtained before its data are used. De-identification is applied at ingestion, access is role-based and audited, and federated training keeps raw data within each hospital. The system is a decision-support tool: it issues no unvalidated clinical decision, defers under uncertainty, and never performs, controls or guides a surgical procedure; all clinical decisions remain the responsibility of the treating clinician (World Health Organization, 2021; European Union, 2024). Fairness is checked by reporting performance by site, age group, sex and tumour type, because bias can arise from apparently reasonable design choices (Obermeyer et al., 2019).

3.12 Technology Stack and Quality Assurance

Models are developed in Python with PyTorch and MONAI (Cardoso et al., 2022); segmentation baselines use nnU-Net and Swin UNETR; pathology uses OpenSlide, QuPath and CLAM with foundation-model encoders; federated learning uses NVIDIA FLARE with Flower as an alternative; surgical planning uses 3D Slicer and the Open Motion Planning Library; the registry uses PostgreSQL, Orthanc and HL7 FHIR. Reproducibility is ensured by experiment tracking, dataset versioning, containerised environments and fixed seeds, so that every reported figure can be regenerated from a recorded model, data and policy version. Code is kept under version control, and progress is recorded on the supervisors' progress website.

3.13 Validity and Reliability

- **Internal validity:** patient-level splits, a held-out test set, pre-declared endpoints and blinded expert review of autonomous cases guard against optimistic estimates.
- **External validity:** leave-one-site-out testing, public external sets and the federated design test generalisation across scanners and institutions.
- **Construct validity:** escalation quality is measured directly (appropriate and missed escalations) rather than inferred from accuracy.
- **Reliability:** double annotation with agreement statistics, repeated runs with different seeds, and versioned artefacts.
- **Qualitative trustworthiness:** an audit trail of coding decisions, triangulation with quantitative escalation data, and member checking with participating clinicians.

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